

Introduction

“A man who keeps company with glaciers comes to feel tolerably insignificant by and by.”

A Tramp Abroad, Mark Twain

1.1 Introduction

Glacier ice covers some 10% of the Earth's land surface at the present time, and covered about three times as much during the ice ages. Currently, all but about 1% of the ice is in areas remote from most human activities, in the great ice sheets of Greenland and Antarctica. Thus it is not surprising that the comparatively small glaciers on mountain areas were the first to attract attention, in both science and literature. Mountain glaciers have long served as natural laboratories for studying glacier processes. They are also important elements of many landscapes; they release water, scour bedrock, cool the weather in summer, and advance down valleys or retreat into high basins. Their spectacular crystalline beauty and conspicuous dynamism captivate observers. Spectacular too are the continental ice sheets, bodies of ice kilometers thick and hundreds of kilometers wide. In contrast to other topographic features of comparable size, ice sheets can change substantially over mere centuries and millennia. They command attention, in part, for their past and future influence on global climate and sea level. They also preserve an extraordinary record of past changes in Earth's climate and atmospheric composition. And the repeated waxing and waning of ice sheets and mountain glaciers throughout the last 2.6 Myr, the Quaternary, has marked vast regions of the Earth's surface with hills and valleys, lakes and river networks, soils and rocky debris – all testimony to the action of flowing ice.

The overall goal of this book is to explain the physical principles underlying the behavior and characteristics of glaciers. *Glaciers*, broadly defined, refers to all ice bodies originating as accumulations of snowfall: mountain glaciers and icefields, small ice caps, continental ice sheets, and floating ice shelves. Study of glaciers is part of glaciology, the study of ice in all its forms. Like other branches of geophysics, glaciology is an interdisciplinary subject involving physicists, geologists, atmospheric scientists, crystallographers, mathematicians, and others. Investigators trained in all these disciplines have contributed in fundamental ways to our understanding of glaciers.

1.2 History and Perspective

Modern glacier studies began in the mid-twentieth century but were founded on a wealth of observations, measurements, and conceptual insights that accumulated since the Enlightenment. The problem of how large, apparently solid masses of ice can flow was studied and debated by many eminent scientists. Altmann, in 1751, correctly recognized that gravity was the cause of glacier motion, but he thought that movement consisted entirely of the ice sliding over its bed. Many glaciers do slide in this way but, in addition, the ice itself can flow like a very viscous fluid, as Bordier suggested in the late eighteenth century. In 1849 Thomson demonstrated ice flow in the laboratory, although the interpretation of his experiment later caused some confusion. Forbes asserted that glacier movement was viscous flow, but Tyndall opposed this view. He thought that motion resulted from the formation of numerous small fractures that subsequently healed by pressure melting and refreezing. Forbes' view prevailed, although only after much heated controversy.

Forbes also observed increases in velocity after heavy rain, which showed that water helps a glacier to slide. The two mechanisms that enable ice to slide past bedrock bumps were identified by Deeley and Parr in 1914. Studies of the flow and storage of water in glaciers date back to dye-tracing experiments by Forel around 1900.

Systematic measurements of glacier flow were begun about 1830 in the Alps. The aim of most early work was to find out how movement varied from place to place on a glacier. Agassiz showed that the velocity is greatest in the central part and decreases progressively toward each side. He also found that a glacier moves more slowly near its head and terminus than elsewhere. Reid, in 1897, showed that the velocity vectors do not parallel the glacier surface; relative to the surface, they are inclined slightly downward in the higher parts of the glacier, where snow accumulates, and slightly upward in the lower reaches to compensate for ice lost by melting. Figure 1.1 illustrates this pattern.

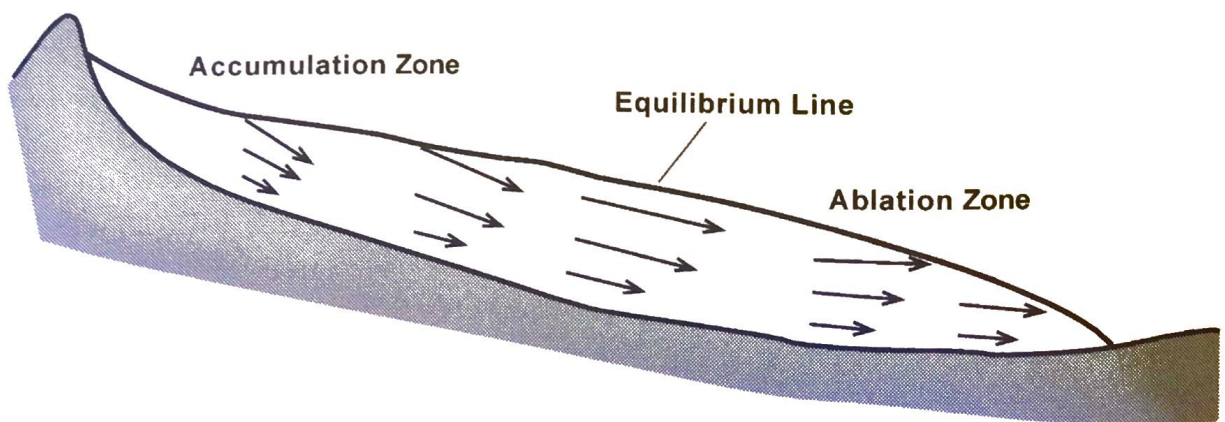


Figure 1.1: Schematic view of velocity vectors in a mountain glacier.

Ice movement at depth was long the subject of debate. *Extrusion flow* – the hypothesis that glaciers flow more rapidly at depth than at the surface – had its proponents, even in the 1930s. In the early 1900s, Blümcke and Hess used a thermal drill in a glacier in the Tyrol and attained bedrock in eleven holes, one of them more than 200 m deep. Rods left in the holes gradually tilted downhill, suggesting that ice moves more rapidly at the surface than at depth. Subsequent borehole measurements using more sophisticated methods have confirmed this pattern in general, although slow across-glacier extrusion flow sometimes occurs in narrow valleys.

Other developments about the turn of the century were the observation by Vallot of a bulge moving down the Mer de Glace and the development of mathematical models of glacier flow by Finsterwalder and others. Reid, for example, analyzed the time lag between the advance of the terminus and the increase in snowfall that produced it. Finsterwalder also pioneered photogrammetric methods of mapping glaciers.

Ahlmann, between 1920 and 1940, carried out classic investigations on the advance and retreat of glaciers in response to changes of climate; he investigated glaciers in Scandinavia, Spitsbergen, Iceland, and Greenland. Complementary studies of how a glacier surface receives heat during the melt season were begun by Sverdrup in 1934.

Work on the ice sheets of Greenland and Antarctica developed more slowly than work on mountain glaciers, but the emphasis of glacier studies has now shifted to the former. Noteworthy early work in Greenland is Koch and Wegener's study of snow stratigraphy during their crossing of the ice sheet in 1913. They also measured temperatures in the ice, in one instance down to a depth of 24 m. Wegener's Greenland Expedition of 1930–1931, which spent the winter in the central part of the ice sheet, studied the transformation of snow to ice. They also made seismic measurements of ice thickness, a method first tried a few years earlier in the Alps. The ice sheet can be regarded as a broad dome, elongated north to south, with a thickness in the center of about 3 km. Close to the edges, much of the ice flows in narrow and fast-moving outlet glaciers along bedrock troughs, a system that was not well delineated until the 1960s. Roughly half of the mass loss occurs by iceberg calving from the fronts of these outlets; the other half, by surface melt around the periphery of the whole ice sheet.

Detailed study of the Antarctic Ice Sheet began with the Norwegian-British-Swedish Antarctic Expedition of 1949–1952. The continent can be divided into three parts (Figure 1.2); the East and West Antarctic Ice Sheets, separated by the Transantarctic Mountains, and the Antarctic Peninsula, home to relatively small ice caps and valley glaciers. Near the coast in both East and West Antarctica, most of the ice flows in outlet glaciers that cut through mountains or in fast-moving ice streams set within comparatively stagnant ice. The outlet glaciers and ice streams feed floating ice shelves that surround much of the continent. Calving of icebergs accounts for most of the mass loss, but melt at the bottom of the shelves contributes significantly. Surface ablation occurs at a few places where mountains block the outflow of ice. Here, winds blow away fresh snow while ice at the surface sublimates, forming areas of "blue ice." Because calving controls most of the mass loss from Antarctica, sea level and ocean temperatures control long-timescale variations in the extent of the ice sheet.

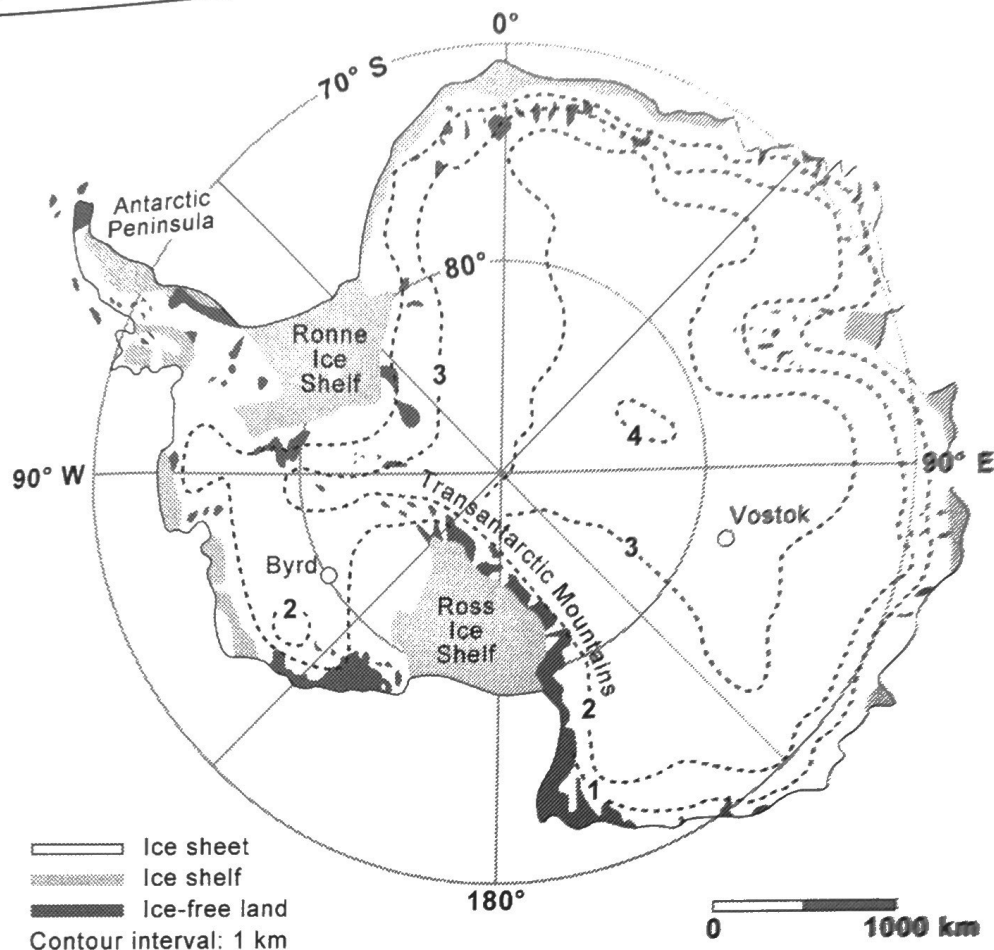


Figure 1.2: Map of Antarctica.

The modern era of glacier studies originated in the 1950s through 1970s with seminal contributions by a handful of mathematical physicists. By integrating physical principles with experimental evidence, their analyses explained many aspects of glacier form, temperature, flow, and evolution (the “glacier dynamics”), and some features of glacial structures and water systems. This body of work continues to underpin nearly all glacier studies. The most important forefront of the science shifted, however, in the 1980s and 1990s to – for lack of a proper term – the geological and climatological interfaces: the complex mix of water, sediment, and irregular bedrock constituting the glacier bed; the equally complex but more readily observed factors controlling melt and accumulation at the glacier surface; and the long and detailed histories of environmental change archived in the ice sheets and studied with ice coring.

This is not to imply that work on fundamentals of glacier dynamics has stagnated since its mid-century surge. On the contrary, glacier dynamics cannot be understood without a proper accounting for processes at the glacier surface and bed. For ice sheets, the long-term history of climate forcings must also be known. Moreover, observational capabilities and computational power have both increased dramatically, allowing analyses to revisit old questions and explore new territory. The application of numerical modelling to glacier problems began in the

1960s and has proliferated with the digital age. A model reduces a complex real situation to a simple closed system that represents the essential features and to which the laws of physics and parameterizations for complex processes can be applied. Modelling can serve three purposes: experimentation, explanation, and prediction. Experimentation, discovering the effect of changing the values of controlling variables, is generally the most useful; it can never be done in the real world. A model explanation may sometimes be illusory; the fact that a model with adjustable parameters gives plausible numerical values does not prove the validity of the underlying assumptions. Most models can be used for prediction, but first they must be tested against data. Unambiguous testing is difficult and use of all the data to “tune” the model by adjusting parameters precludes a proper assessment of its abilities. Events in the past decade gave a harsh reminder of the need to maintain skepticism about models; ice sheet models did not predict the dramatic recent increase of ice outflow from both Greenland and Antarctica. Although they are sophisticated and useful for a variety of studies, these models do not yet simulate nature adequately.

Many types of glacier studies continue to yield important results, but, above all else, the past decade has been a period of unmatched progress for observational analyses of the polar ice sheets. With data from remote sensing, the entire outflow of ice from Antarctica has been measured within a single year. So too for Greenland. Changes of surface elevation have been mapped throughout broad regions. Measurements of subtle variations in the strength of gravity, using satellites, now provide estimates of year-to-year and even seasonal variations in total mass of both ice sheets. Nearly complete maps of the velocity and thickness of ice streams permit a rigorous, if uncertain, assessment of all the forces acting on them. Arrays of shallow ice cores now show in unprecedented detail the geographical patterns of snow accumulation. With this information, atmospheric models have been calibrated and then used to map the year-to-year variations of mass gain or loss at the surface. Until recently, none of these analyses was feasible.

These capabilities, among others, will play an essential role in answering the major glaciological question of our time: how will ice on Earth respond to widespread climatic warming? At the very least, glacier changes can be monitored as they happen. At best, predictions of future changes will rigorously delimit the range of plausible behaviors and elaborate the most likely ones. Precise predictions seem impossible; even if glaciers were completely understood and characterized, the uncertainty of climate forecasts would still allow for a wide range of scenarios.

The evolution of the world’s glaciers, past and future, depends on geological and climatological contingencies such as the distribution of subglacial sediments, the shapes of bedrock troughs beneath ice streams, and the interaction of mountain ranges with atmospheric winds and precipitation. Appreciation for the powerful insights in the seminal works of glaciology sometimes leads to a fetishizing of “general principles” as the goal of the science. Though invaluable, general principles by themselves do not answer the questions that matter. The fate of the ice sheets in a warming climate, for example, depends on the outflow along a few dozen outlet glaciers and ice streams in Greenland and Antarctica. These features are surprisingly diverse.

Each one needs to be understood and, ultimately, represented in models of each ice sheet. Again, the potential for melt and break-up of each major ice shelf – and how such changes relate to conditions in the nearby ocean – depends on particularities of the subshelf waters and ocean floor, the shape of the enclosing embayments, and other factors.

One particularity of great significance concerns the difference between the two ice sheets in mainland Antarctica (Figure 1.2). Although the greatest ice thickness in East Antarctica is nearly 4.8 km, most of the bedrock is above sea level. Much of the base of the West Antarctic Ice Sheet, in contrast, sits well below sea level and would remain so after isostatic rebound following removal of the ice. This suggests a possible instability; thinning of the ice past a certain level could decouple parts of the ice sheet from the bed, initiate rapid outflow, and soon set the remaining ice afloat.

Just as it is essential to match theoretical work with a realistic view of such contingencies and complexities in nature, it is essential that experimental work be coordinated with theory. Experiments should be designed to solve specific problems or characterize important components of a system. The mere acquisition of data is seldom a useful contribution in itself.

Understanding of glaciers has improved dramatically over the past half-century or so. Modern techniques of measurement would surely astonish Forbes and Agassiz, as would the abundance of information now available to evaluate theoretical models. Yet the deep inadequacies of the science should be kept in mind. Observation of the glacier bed remains a very difficult task. Interpretations of such data are usually tenuous, and theoretical views of glacier bed processes do a poor job of anticipating important new observations. Drilling boreholes in ice sheets is expensive and time-consuming; it is rarely done for dynamics studies. Of the dozens of fast-flowing polar ice streams, only one group in one sector of Antarctica has ever been studied by boreholes reaching the bed or even penetrating the deep layers of ice. Keeping track of year-to-year variations in melt and snowfall on “representative” mountain glaciers requires a consistent commitment of resources that faces vexing difficulties even in the wealthiest nations, let alone nations experiencing political upheavals. Geological reconstructions of past ice incursions provide the best hope for validating numerical models – but such studies must overcome imprecision and ambiguity. In short, much remains to be learned about glaciers, and even maintaining the current level of observation will take concerted effort.

1.3 Organization of the Book

The various aspects of glacier studies intersect one another in numerous ways, and any choice of organizational scheme is therefore somewhat arbitrary. The following summary clarifies our intentions.

Glacier ice forms by compaction and metamorphism of snow, sometimes accompanied by melting and refreezing. Chapter 2 discusses this process and the basic materials – snow, firn, and ice – that compose a glacier. One motivation for studying the transformation of snow to ice is to understand what determines the variation of density with depth near the surface of polar ice